Traded-Coordinate Diagnostics for Hedging Feedback

A self-contained route from market episodes to the theorem

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Talk route

Market feedback episodes → clearing resolvent → pathwise P&L decomposition → Grassmannian obstruction.

Market anchors: one repeated pattern

Core motifs

- **1987 portfolio insurance:** dynamic hedging sells into falling markets.
- **Feb. 2018 inverse-vol / vol-control:** rebalancing flow makes exposure endogenous.
- **VaR, margin, target leverage:** risk limits force deleveraging into stress.
- **Autocallable / ELS books:** barrier-localized Greeks concentrate dealer hedging.

Not an empirical claim

These episodes anchor the mechanism; they are not calibration inputs. The paper abstracts one loop:

$$\text{price move} \to \text{hedge/risk-rule update} \to \text{flow} \to \text{impact} \to \text{price move}$$

Question

Can we separate genuine feedback P&L from ordinary model misspecification?

The model in one picture

Two traded-coordinate panels

- $\widetilde{Z}$: latent no-impact local martingale.
- $\widetilde{Y}$: impacted observed panel, e.g. $Y = (S, C^1, \dots, C^r, P^1, \dots, P^s)$.

Memoryless clearing:

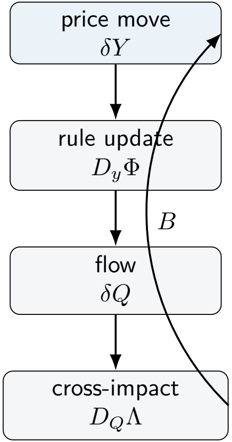
$$Y = Z + \Lambda(\Phi(Y)),$$

$$B = D_Q\Lambda\, D_y\Phi, \qquad J = (I - B)^{-1}.$$

$D_Q\Lambda$ is full cross-impact: flow in one instrument can move the whole panel.

Φ is supplied by the audited policy: Greeks in the paper's barrier case; balance-sheet/VaR sensitivity in leverage loops.

Clearing is a local fixed point with supplied Φ, not a strategic/MFG optimization.



$$\boxed{\delta Y = J\,\delta Z}$$

Why pathwise reasoning is the right language

Layer-1 bookkeeping between switching times

For discounted wealth and mark $v(t, y, \mathfrak{I})$,

$$\widetilde{X}_T - v_T = \widetilde{X}_0 - v_0 - \int_0^T \mathfrak{r}_t \, \mathrm{d}t - \frac{1}{2} \int_0^T H_t : (c_t^Y - \widehat{c}_t) \, \mathrm{d}t.$$

Why pathwise

- Fix ω on the full-measure set where the semimartingale identities hold.
- Quadratic covariation is pathwise for continuous semimartingales.
- Equivalent measure changes do not alter c^Y or c^Z.
- Hedging drift cancels through self-financing bookkeeping.

Scope

The theorem is continuous-semimartingale and non-fold. Jumps, local time at nonsmooth barriers, and through-fold dynamics require extra terms.

The algebra: dual congruence

The invariant pairing

$$g \cdot (H, C) = \big(g^{-\top} H g^{-1},\ g C g^{\top}\big), \qquad \operatorname{tr}\big(g^{-\top} H g^{-1} g C g^{\top}\big) = \operatorname{tr}(HC).$$

Apply it to clearing

Clearing supplies the group element $g = J^{-1} = I - B$. Differentiating the clearing branch gives

$$c^Y = J c^Z J^{\top}, \qquad J = (I - B)^{-1}.$$

The Hessian leg transforms dually:

$$H : c^Y = (J^{\top} H J) : c^Z.$$

Takeaway

The observed covariance is not merely noisy. Under feedback it is a congruence transform of a no-impact counterfactual.

Main decomposition: feedback versus benchmark error

Unified pathwise identity

$$\underbrace{-\frac{1}{2}\int H:(c^Y-\widehat{c})\ \mathrm{d}t}_{\text{total carry error}} = \underbrace{-\frac{1}{2}\int (J^\top HJ-H):c^Z\ \mathrm{d}t}_{F\text{: feedback carry}} + \underbrace{\frac{1}{2}\int H:(\widehat{c}-c^Z)\ \mathrm{d}t}_{M\text{: benchmark miss}}.$$

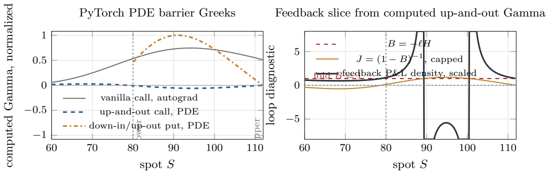
Desk interpretation

F is generated by the desk's own loop gain B. It vanishes when feedback does not alter the Hessian-covariance pairing.

Identification limit

$c^Z=(I-B)c^Y(I-B)^\top$ is observable after B is calibrated, but the split $F+M$ is not identified by $(H,c^Y,\widehat{c})$ alone.

Where barriers amplify feedback



Computed PDE barrier gammas and scalar feedback slice.

The bridge

For a delta hedge,

$$\Phi = -\nabla_y v, \qquad D_y\Phi = -H, \qquad B = -LH.$$

Barriers matter because H is large and localized. The feedback question is whether $I - B$ remains safely invertible.

Normal-impact special case

If $L = L^\top \succ 0$,

$$B \sim -L^{1/2} H L^{1/2}.$$

Then only negative Hessian directions can reach $\det(I - B) = 0$ as impact scale grows. For nonsymmetric cross-impact, inspect $B = D_Q \Lambda D_y \Phi$ directly.

Volterra impact: memory behind the local Jacobian

Causal transient impact primitive

$$Y_t = Z_t + M_t, \qquad M_t = \int_0^t K(t-s)\big(\Phi(s, Y_s, \mathfrak{I}_s) + \dot{Q}_s^{\mathrm{ext}}\big)\,\mathrm{d}s.$$

Why memory is introduced

- real impact is causal and resilient;
- the kernel calibrates the slope $D_Q\Lambda$;
- finite-memory stress tests live here, not in the static formula.

Why exponential here

For $K(u) = \kappa e^{-\kappa u} L$:

$$\dot{M} = -\kappa M + \kappa L(\Phi(Y) + \dot{Q}^{\mathrm{ext}}).$$

Fast resilience gives the memoryless branch

$$Y = Z + L(\Phi(Y) + \dot{Q}^{\mathrm{ext}}), \qquad B = L D_y \Phi.$$

Scope

The core theorem uses the memoryless branch. Permanent impact is a separate fundamental or cumulative state, not this transient feedback term.

No fold, no bifurcation, no oscillation: what is actually proved

Non-fold is the theorem's branch condition

$$I - B_t \in \mathrm{GL}(n+1).$$

It ensures a local clearing branch and the covariance conjugation. No uniform resolvent bound is imposed; contraction $\|B\| < 1$ is only a sufficient stability certificate. Diagnose with $\sigma_{\min}(I - B)$ and $\|J\|_{\mathrm{op}}$, not $\rho(B)$: a non-normal B can have spectral radius below 1 yet a large resolvent.

No bifurcation claim

The paper stops before $\det(I - B) = 0$. Through-fold dynamics need singular perturbation and a global vector field; they are future work.

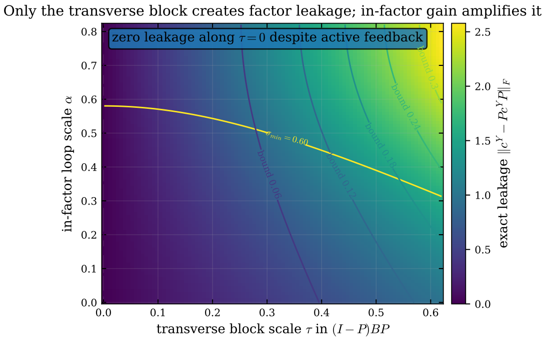
Why no oscillation here

Finite-κ linearization:

$$\dot{\delta} = -\kappa(I - B)\delta.$$

With potential/normal impact and delta hedge, $B = -LH$ has real eigenvalues. Thus one exponential mode gives damping or monotone instability, not an autonomous oscillator. Oscillation needs non-symmetric cross-impact, multiple kernels, or reaction lag.

Why Grassmannian geometry appears



Transverse feedback creates structural leakage.

What recalibration can change

$$\mathcal{M}(G) = \{G\Sigma G^\top : \Sigma \in \mathbb{S}^d_{++}\}.$$

G: factor directions in the panel Y; $\mathrm{col}(G)$: moves the model can represent. Σ: factor volatilities/correlations *inside* that span.

What feedback can change

If $c^Z = G\Sigma^Z G^\top$, then

$$c^Y = Jc^Z J^\top = (JG)\Sigma^Z (JG)^\top.$$

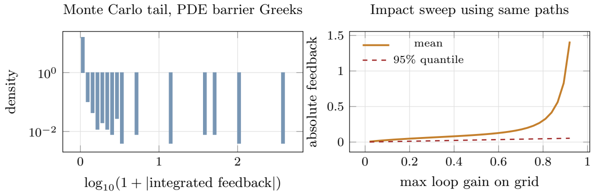
Feedback replaces G by JG. It is only a Σ-recalibration iff

$$J\,\mathrm{col}(G) = \mathrm{col}(G) \iff (I - P)BP = 0, \quad P = \mathrm{proj}_{\mathrm{col}(G)}.$$

Why Grassmannian

$\theta_k = \angle_k(\mathrm{col}(G), J\,\mathrm{col}(G))$ are principal angles: all zero means the same factor space; any positive angle is off-factor leakage that Σ cannot absorb.

Barrier simulation: real computation



PyTorch PDE Greeks, 4 096 paths, 252 steps.

Implementation

- Crank–Nicolson PDE grids for barrier values.
- Autograd and finite differences for Greeks/Hessians.
- Monte Carlo paths evaluated with computed Greeks.

Practical finding

Barrier feedback is tail-concentrated: near-boundary paths dominate absolute feedback carry. Mean volatility diagnostics can miss this.

Operational method map: no hidden step

1. Market data
$Y = (S, \text{calls}, \text{puts})$, flows, contracts, P&L; estimate c^Y.

2. Risk engine
Compute $v, H, \Phi, D_y\Phi, \widehat{c}$ by PDE, MC, AAD, or differences.

3. Impact leg
Estimate full $D_Q\Lambda$; optional kernels fit resilience.

4. Loop and clearing
$B = D_Q \Lambda D_y \Phi$; check $m = \sigma_{\min}(I - B)$; compute J, c^Z.

5. P&L split
Contract H with $c^Y, c^Z, \widehat{c}$ to get feedback F and miss M.

6. Factor obstruction
Estimate G; QR/SVD gives P; report leakage and principal angles.

Read this as an algorithm

The empirical hinge is $D_Q\Lambda$; once $H, c^Y, \widehat{c}$, and B are fixed, the feedback/misspecification split is algebraic.

Future research horizon

Mathematical frontier

- Through-fold analysis: loss of normal hyperbolicity.
- Operator resolvents for general Volterra kernels.

Finance frontier

- Optimal hedging when $J = (I - D_Q \Lambda D_y \Phi)^{-1}$ depends on control.
- Aggregate loop gain $B^{\mathrm{agg}} = \sum_k B^{(k)}$: cross-desk words $B^{(i)} B^{(j)}$ amplify, and aligned books reach a synchronized fold.
- Filtering and impact-calibration uncertainty.

Closing sentence

The paper turns feedback hedging into an auditable pathwise identity: compute B, check the non-fold margin, separate F from M, and test whether the factor span survives the clearing resolvent.